

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
MS by Research in Computer Science and Engineering	International Institute of Information Technology, Hyderabad	8.00	Jan 2026 - Present
B.S. Data Science and Engineering	Indian Institute of Science Education and Research, Bhopal	8.41	Dec 2021 - May 2025
Senior Secondary	CBSE Board	90.0%	2021
Secondary	CBSE Board	95.0%	2019

EXPERIENCE

- Centre for Visual Information and Technology (CVIT), IIIT Hyderabad

Applied Research Fellow

May 2025 - Dec 2025

Offer Letter
- Designed RoadTones: a dataset-model-evaluation stack, which enables fine-grained, tone-controllable road-video captioning. Work accepted at CVPR Findings 2026 ([Project page](#)).
- Indian Institute of Technology Madras

Research Intern

May 2024 - Jul. 2024

Certificate
- Explored Super-Resolution techniques for 3D medical imaging, enhancing z-axis resolution in CT and MRI scans.

– Utilized GANs, Diffusion Models, and NeRF to improve image clarity for better diagnostics: [Final Report](#)

PROJECTS

- BS Thesis - Image Tampering Detector

IISER Bhopal

Jan. 2025 - April 2025

Report
- Developed an advanced segmentation approach to enhance tampered region detection by improving transformer-based methods such as SAFIRE and DocTamper.

– Integrated Discrete Wavelet Transform (DWT) as a superior alternative to Discrete Cosine Transform (DCT), effectively capturing both frequency and spatial domain features, resulting in more precise localization of tampered regions.

– Enhanced model efficiency by achieving higher accuracy with significantly fewer training epochs, showcasing the impact of frequency-domain analysis in forensic image segmentation while optimizing computational performance.
- Improving Monocular Depth Estimation of outdoor images in Dim Light Conditions

IISER Bhopal

Sept. 2024 to Nov. 2024

Report
- Integrated denoising and deblurring filters into the transformer-based DepthAnything V2 model to improve depth estimation in blurry and noisy nighttime images.
- Prediction of Mortality Rate of Heart Failure Patients Admitted to ICU

IISER Bhopal

Sept. 2023 - Nov. 2023

GitHub
- A Binary Classification problem that has been executed using Supervised Machine Learning techniques. Several classifiers and pre-processing methods were applied to check which classifier gave the best results.

TECHNICAL SKILLS

- **Programming Languages:** Python3, C/C++, JavaScript, MATLAB, R, HTML, CSS, MySQL
- **Technologies:** PyTorch, Tensorflow, Keras, OpenCV
- **Other:** Docker, Firebase, Android SDK, Data Structures and Algorithms, Linux

KEY COURSES TAKEN

- **Mathematics:** Linear Algebra, Multivariable Calculus, Discrete Mathematics, Probability and Statistics
- **Data Science and Engineering:** 3D Deep Learning, Natural Language Processing (NLP), Data Analyses and Statistics for Geosciences, Applied and Accelerated AI, Deep Learning, Artificial Intelligence (AI), Machine Learning, Computer Vision, Database Management Systems, Applied Optimisation, Econometrics, Data Science in Practice (Advanced Python), Data Structures and Algorithms, Signals and Systems

ACHIEVEMENTS

- **Publication**, 'RoadTones: Tone Controllable Text Generation from Road Event Videos' accepted at the CVPR 2026 Findings Track. *2025*
- **Presentation**, Poster Titled 'Improving Image Tampering Detection: A Hybrid Approach Using Frequency and Spatial Domain Features' presented at CV4Science Workshop, CVPR 2025 *2025*
- **GATE DA (Data Science and Artificial Intelligence)**, AIR - 1455, Score - 534 *2025*
- **ISRO Robotics Challenge**, Among Top 30 teams out of 400 in India *2024*
- **IEEE CAS Student Design Competition**, 3rd place in Asia Pacific Region *2022*
- **JEE Mains**, 97.195 Percentile *2021*
- **IISER Aptitude Test**, AIR - 1513 *2021*